

Architecture

- **System Integration:** The engineering practice of combining disparate software subsystems (File I/O, Cloud AI, Browser Automation) into a singular, cohesive operational pipeline.
- **The Orchestration Pattern:** A centralized architectural design where a master script controller (`main.py`) acts as a strict programmatic conductor, managing application state and routing data between isolated worker modules.
- **Volatile Memory Data-Passing:** The technique of holding extracted data inside temporary system RAM variables (like our `structured_application_data` dictionary) to transition it smoothly between backend logic and frontend user interfaces without needing to write intermediate files to disk.
- **Deterministic Pipeline Guards:** Code logic switches (like `sys.exit(1)`) placed between critical data integration steps to immediately halt operations if a dependency component fails, preventing downstream errors.